

# Recall: SFT Requires Teacher Solutions!

Problem Prompt ( $X$ )

Teacher Model Solution

Teacher Solution per token



$Y_1$

$Y_2$

$Y_3$

$Y_4$

...

$Y_T$

Loss

$$-\log P(Y|X) = -\sum_{t=1}^T \log P(Y_T|X, Y_{<t})$$

Student Probability per token



$P(Y_1|X)$

$P(Y_2|X, Y_1)$

...

$P(Y_T|X, Y_{<T})$

Probability of generating teacher sequence



